Paper 3 Analysis Table

Category	Details
Research Paper Title	*DeepBreastCancerNet: A Novel Deep Learning Model for Breast Cancer Detection Using Ultrasound Images*
Objective	To propose a **new deep learning model (DeepBreastCancerNet)** for accurate breast cancer detection and classification from ultrasound images.
Problem Statement	Manual detection of breast cancer using ultrasound is **time-consuming, complex, and prone to misdiagnosis**, which can lead to **unnecessary surgeries and incorrect treatment**.
Model Used	- Proposed Model: **DeepBreastCancerNet** - 24 layers (CNN-based) - Includes convolution layers, inception modules, fully connected layer
Programming / Tools	➤ Python ➤ Deep Learning frameworks (TensorFlow / PyTorch) ➤ Transfer Learning ➤ Data Augmentation
Dataset	- BUSI Dataset (Kaggle) - 780 images: ➤ Benign (437) ➤ Malignant (210) ➤ Normal (133)
Pros	✔ Very high accuracy (up to **99.63%**) ✔ Novel architecture ✔ Uses data augmentation ✔ Compared with multiple DL models ✔ Strong performance improvement over existing methods
Cons	✘ Small dataset size ✘ Risk of overfitting (especially with transfer learning) ✘ No explainability (black-box model) ✘ Limited real-world validation

Challenges	⚠ Limited dataset availability ⚠ Class imbalance (fewer malignant images) ⚠ Ultrasound noise & low contrast ⚠ Computational complexity of deep models
Conclusion	The proposed **DeepBreastCancerNet model outperforms existing CNN models** and achieves very high accuracy for breast cancer detection using ultrasound images.
Critical Comment	This paper is very strong in **model performance (accuracy ~99%)**, but like previous papers, it **completely lacks Explainable AI (XAI)**. For your FYP (*UltraXAI*), this is a key opportunity: you can take such high-performing models and integrate **Grad-CAM or other XAI methods** to make predictions interpretable for doctors. Also, dataset size and imbalance remain major limitations — your project can improve by using **better preprocessing, augmentation, or multiple datasets**.